

TABUK, Saudi Arabia

WMO: 403750

Lat: 28.365N

Lon: 36.619E

Elev: 2551

StdP: 13.39

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	35.8	38.8	10.0	10.1	60.2	13.8	12.2	61.0	26.1	65.1	22.7	65.0	2.4	110	0.438

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	26.1	107.3	66.0	104.4	65.1	102.4	64.5	69.6	97.6	68.0	96.0	66.7	95.8	9.0	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
59.4	83.1	80.8	57.4	77.4	80.8	55.6	72.3	79.9	35.3	97.6	33.7	96.2	32.7	97.6	85.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.6	19.0	15.8	31.4	110.1	2.7	1.9	29.5	111.4	27.9	112.6	26.4	113.7	24.4	115.1
			26.7	70.9	2.6	3.7	24.9	73.6	23.4	75.7	21.9	77.8	20.0	80.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	72.9	53.0	57.2	64.0	72.8	80.8	86.4	89.4	89.8	85.0	76.5	64.2	55.1	(6)
	DBStd	14.11	5.51	6.11	6.60	6.31	5.18	3.51	3.31	3.40	3.76	5.24	5.73	5.28	(7)
	HDD50	46	28	7	0	0	0	0	0	0	0	0	0	11	(8)
	HDD65	1107	374	229	100	11	0	0	0	0	0	1	83	309	(9)
	CDD50	8415	121	209	435	683	954	1092	1220	1234	1049	822	426	170	(10)
	CDD65	4001	2	11	70	244	489	642	755	769	599	358	58	4	(11)
	CDH74	56313	45	187	942	2951	6421	9424	11687	11835	8170	3989	600	62	(12)
	CDH80	33713	3	38	320	1360	3649	5929	7601	7747	4944	1963	152	7	(13)
(14) Wind	WSAvg	6.3	5.1	6.2	6.9	7.4	7.6	7.4	7.1	7.1	6.2	5.5	4.7	4.4	(14)
(15) Precipitation	PrecAvg	1.2	0.4	0.0	0.2	0.0	0.1	0.0	0.0	0.0	0.0	0.1	0.1	0.1	(15)
	PrecMax	3.8	3.0	0.5	1.0	0.5	2.0	0.0	0.0	0.1	0.1	0.8	1.6	0.7	(16)
	PrecMin	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
	PrecStd	0.9	0.7	0.1	0.3	0.1	0.4	0.0	0.0	0.0	0.0	0.2	0.3	0.2	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	78.9	84.9	93.0	98.5	104.1	107.6	109.6	109.6	107.0	99.3	88.2	80.4	(19)
		MCWB	55.2	56.6	58.5	61.0	63.1	64.2	65.9	67.2	65.4	62.7	58.8	55.9	(20)
	2%	DB	73.7	79.5	87.8	94.9	100.5	104.2	106.2	107.4	102.6	96.5	84.3	75.0	(21)
		MCWB	52.8	54.5	56.7	60.5	62.2	63.9	65.8	67.0	64.6	62.0	57.8	54.0	(22)
	5%	DB	69.7	75.2	82.8	91.5	98.2	102.0	104.0	104.3	100.4	93.3	80.7	71.2	(23)
		MCWB	51.3	52.5	55.4	58.5	61.5	63.3	64.8	66.1	64.0	61.0	56.8	52.6	(24)
	10%	DB	65.4	71.2	79.0	87.9	95.1	100.0	101.8	102.2	98.2	90.0	77.1	67.5	(25)
		MCWB	49.5	51.0	53.8	57.3	60.4	62.9	64.3	65.6	63.5	60.4	55.8	51.1	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.2	59.7	61.6	65.3	68.3	70.1	71.2	72.3	70.3	67.0	62.8	60.1	(27)
		MCDB	71.3	75.3	83.8	91.7	94.1	99.2	97.7	100.7	95.8	89.3	78.4	72.3	(28)
	2%	WB	55.8	56.9	58.9	62.2	64.9	67.2	68.7	69.8	67.4	64.4	60.6	56.5	(29)
		MCDB	68.5	74.8	82.4	87.5	91.1	95.6	96.7	99.0	94.0	87.4	76.6	71.3	(30)
	5%	WB	52.6	54.3	56.7	60.2	62.8	65.1	67.1	68.1	65.7	63.0	58.8	53.8	(31)
		MCDB	67.1	72.1	79.3	87.0	92.7	95.3	97.0	96.9	93.2	86.5	74.5	67.5	(32)
	10%	WB	50.1	51.9	54.8	58.4	61.4	63.6	65.6	66.8	64.8	61.7	57.2	51.7	(33)
		MCDB	64.0	69.0	76.8	85.5	91.5	94.9	96.8	96.5	92.4	85.3	73.9	65.7	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	24.0	25.1	26.1	27.0	26.9	26.9	25.4	26.1	26.8	26.2	24.3	24.3	(35)
		MCDBR	29.0	30.3	30.2	29.9	28.9	28.6	27.0	27.5	28.5	29.0	28.2	27.9	(36)
	5% WB	MCWBR	15.0	14.6	13.0	12.1	10.6	9.5	8.2	8.5	9.2	10.7	12.2	14.2	(37)
		MCWBR	25.5	26.6	27.8	27.6	27.1	27.4	25.7	26.5	26.7	25.8	23.4	24.7	(38)
(40) Clear-Sky Solar Irradiance	taub		0.307	0.338	0.390	0.462	0.456	0.397	0.408	0.422	0.395	0.398	0.347	0.313	(40)
		taud	2.401	2.292	2.109	1.914	1.960	2.165	2.164	2.123	2.178	2.152	2.297	2.394	(41)
	Ebn at Noon		297	295	284	266	267	282	279	274	279	270	279	289	(42)
		Edh at Noon	32	38	49	62	59	48	48	49	45	44	35	30	(43)
(44) All-Sky Solar Radiation	RadAvg		1325	1635	2021	2296	2483	2666	2563	2385	2127	1740	1397	1231	(44)
	RadStd		61	70	55	55	96	30	54	51	59	57	55	30	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	+1.12	N/A	-3.49	+1.44	N/A	N/A	N/A	-133	+434	+320	
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page